Community structure of the pseudofractal web

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The Ramsey community number r_κ is the smallest network size at which a graph is better described by a partition into communities than by no partition, under a prescribed detection rule. On a scale-free graph this question is confounded: a block model can split the network merely to absorb its degree distribution. I compute r_κ *analytically* for the deterministic pseudofractal scale-free web of Dorogovtsev, Goltsev, and Mendes, separating genuine community structure from degree heterogeneity with two closed-form detection rules. Under a plain Bernoulli stochastic block model, the web's natural recursive bipartition is unpreferred while small and breaks at $r_\kappa = 1095$ nodes, with a log-evidence growing as $(\ln 3 - \frac{2}{3}\ln 2)n$. Under a degree-corrected model tested against the configuration-model null, the same partition survives, breaking far earlier at $r_\kappa = 42$, with a log-evidence growing as $(2\ln 3 - \frac{4}{3}\ln 2)n$— exactly twice the plain slope, and independent of the prior. Degree correction reverses the ordering of the candidate cuts, demoting the hub–leaf split and elevating the recursive one. Because the web is self-similar, the best description is not two communities but a nested hierarchy: the degree-corrected evidence keeps rising as the partition is refined, and is maximised at of order $\sqrt{n}$ communities of $\sim\sqrt{n}$ nodes. A purely local recursive rule thus builds true hierarchical community structure, over and above the scale-free degree sequence it also produces, in an exactly solvable setting.

I. INTRODUCTION

Community structure—the organisation of nodes into groups more densely connected internally than to the rest of the network—is one of the most studied features of complex networks [1–3]. It is usually attributed to node heterogeneity, but node heterogeneity is not a necessary condition: networks grown by purely local rules [4] routinely acquire communities as they grow [5]. To make this quantitative, Ref. [5] introduced the *Ramsey community number* r_κ, the minimum graph size that guarantees, with near-certainty, that a prescribed detection method reports two or more communities; the name evokes Ramsey theory, in which a large enough structure unavoidably contains ordered substructures [6]. A companion paper computed r_κ analytically for the circulant ring lattice $C_n(1,\ldots,c)$, a homogeneous locally wired graph, using a Bernoulli stochastic block model (SBM) with symmetric Beta priors as the detection rule: the plain cycle ($c = 1$) is never partitioned ($r_\kappa = \infty$), while the next-nearest neighbour ring ($c = 2$) acquires a finite r_κ of a few tens of nodes, with a log-evidence growing as $(\ln 2)\, n$ [**?**].

Those results rest on a special feature of the ring: it is *degree regular*. In a regular graph a two-block partition can only capture assortative community structure, because there is no degree variation to be explained away. Most networks of interest are not regular, and the sharpest test of the emergence idea is a graph that is strongly heterogeneous, where the naive question "does it want to split?" becomes confounded. On a scale-free graph a plain block model will happily split the network into a high-degree core and a low-degree periphery, not because the two groups form communities but simply because they have different mean degrees. Distinguishing genuine community structure from this degree artefact is the central issue of degree-corrected community detection [7].

Here I address both questions at once, analytically, for a canonical deterministic scale-free graph: the pseudofractal web of Dorogovtsev, Goltsev, and Mendes [8]. This graph is built by a fixed recursive rule, has a power-law degree distribution with exponent $\gamma = 1+\ln 3/\ln 2 \simeq 2.585$, and—crucially for an exact treatment—has a natural recursive bipartition whose block sizes and edge counts are known in closed form at every generation. I compute the Bayesian evidence for this partition under two detection rules. The first is the plain Bernoulli SBM. The second is a degree-corrected Poisson SBM tested against the configuration-model null, which holds the degree sequence fixed and asks whether the partition explains structure *beyond* what the degrees already imply. Both evidences are single closed-form expressions, so the transition and r_κ follow exactly.

The answers are clean and complementary. Under the plain SBM the web is unpartitioned while small and breaks at $r_\kappa = 1095$ nodes, with $\log R \sim (\ln 3 - \frac{2}{3}\ln 2)n$. Under degree correction the same partition breaks much earlier, $r_\kappa = 42$, with a log-evidence growing exactly twice as fast, $\log R_{\rm dc} \sim (2\ln 3 - \frac{4}{3}\ln 2)n$, and both slopes are independent of the prior. Degree correction reverses the ordering that the plain model assigns to the competing cuts: the hub–leaf split that the plain model most prefers is demoted, and the recursive split is elevated. Pushing this further, the self-similarity of the web makes the optimal description a nested hierarchy of communities, refined down to $\sim\sqrt{n}$ blocks. The recursion therefore builds true hierarchical community structure on top of its scale-free degree sequence. The rest of the paper de-

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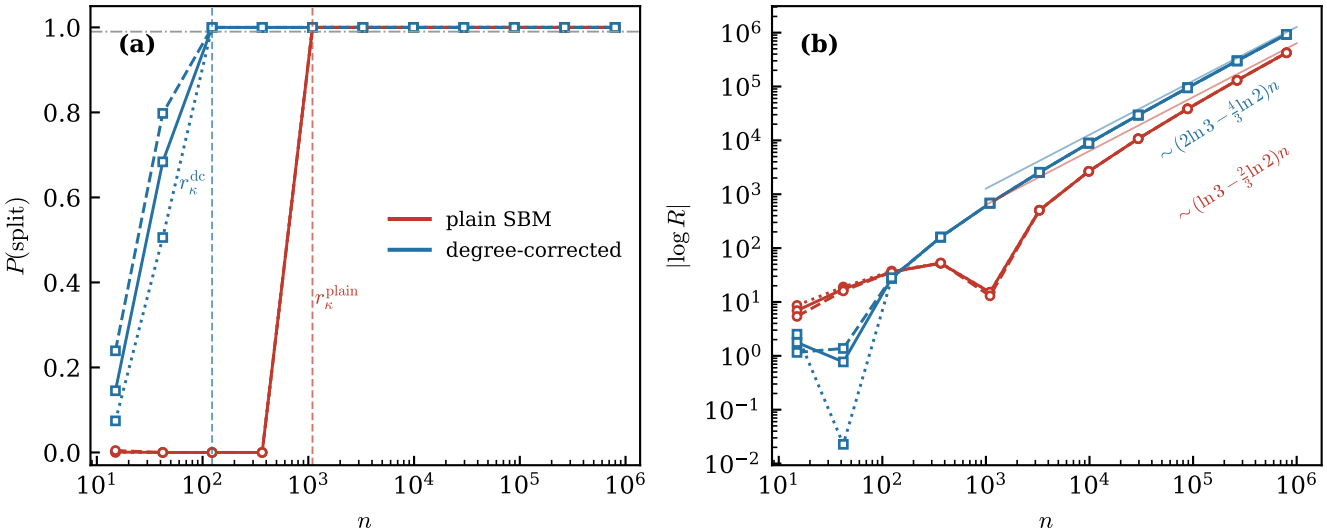

FIG. 1. (a) Posterior weight of the recursive branch partition, Eq. (7), versus network size n (log scale). Red: plain Bernoulli SBM; blue: degree-corrected (config-null) model. Line styles denote $\alpha = 0.5$ (dotted), 1 (solid), 2 (dashed). The dash-dotted line marks $q = 0.99$; vertical guides mark the Ramsey community numbers $r_\kappa^{\rm plain} = 1095$ and $r_\kappa^{\rm dc} = 123$ at that certainty. (b) The magnitude of the log-evidence ratio, $|\log R|$; both models grow linearly in n, with prior-independent slopes $\ln 3 - \frac{2}{3}\ln 2$ (plain) and $2\ln 3 - \frac{4}{3}\ln 2$ (degree-corrected), the latter exactly twice the former.

fines the web and the partition (Sec. II), derives the plain evidence ratio and its transition (Sec. III), constructs the degree-corrected evidence and shows the partition survives it (Sec. IV), resolves the hierarchical community structure (Sec. V), reports the Ramsey community numbers (Sec. VI), and concludes (Sec. VII).

II. MODEL AND DETECTION RULE

A. The pseudofractal web

The pseudofractal scale-free web [8] is grown deterministically. Start ($t = 0$) from a triangle: three vertices and three edges. At each subsequent generation, every existing edge spawns one new vertex, joined to both endpoints of that edge; old edges are retained. Counting gives, at generation t,

$$n_t = \frac{3}{2}\big(3^t + 1\big), \qquad E_t = 3^{\,t+1}, \tag{1}$$

so the mean degree $2E_t/n_t \to 4$ and the graph is sparse. A vertex introduced at generation τ has its degree doubled at every later generation, so degrees take the values $2^{\,t-\tau+1}$ and their distribution is a power law $P(k) \sim k^{-\gamma}$ with $\gamma = 1 + \ln 3/\ln 2 \simeq 2.585$. The three seed vertices are permanent hubs of maximal degree $2^{\,t+1}$.

B. The recursive branch bipartition

Every non-seed vertex descends from exactly one edge, which in turn belongs to one of the three "branches" rooted on the three sides of the initial triangle. This assigns a branch label $b \in \{0, 1, 2\}$ to each of the $n_t - 3$ non-seed vertices, $(3^t - 1)/2$ per branch, while the three seed hubs are shared. The natural recursive bipartition places one branch together with the three hubs in block 1, and the other two branches in block 2; it respects the $\mathbb{Z}_2$ symmetry exchanging the two branches of block 2. Writing $T \equiv t$ for brevity, the block sizes and the within/between edge counts are all closed form (verified against direct construction through $T = 8$):

$$n_1 = \tfrac{1}{2}\big(3^T + 5\big), \qquad n_2 = 3^T - 1, \tag{2}$$

$$E_1 = 3^T + 2, \qquad E_2 = 2\cdot 3^T + 2 - 2^{\,T+2}, \tag{3}$$

$$E_{12} = 2^{\,T+2} - 4, \tag{4}$$

with $n_1 + n_2 = n_T$ and $E_1 + E_2 + E_{12} = E_T = 3^{T+1}$. The seam between the two blocks is thin: $E_{12} = 2^{T+2} - 4$ grows only as $2^T = n^{\ln 2/\ln 3} \simeq n^{0.631}$, so the fraction of crossing edges $E_{12}/E_T \sim (2/3)^T \to 0$. The two branches are joined by a vanishing density of edges (Fig. 2)—the structural fact that drives everything below.

C. Stochastic block model evidence

The detection rule is Bayesian model comparison between the bipartition of Sec. II B and the

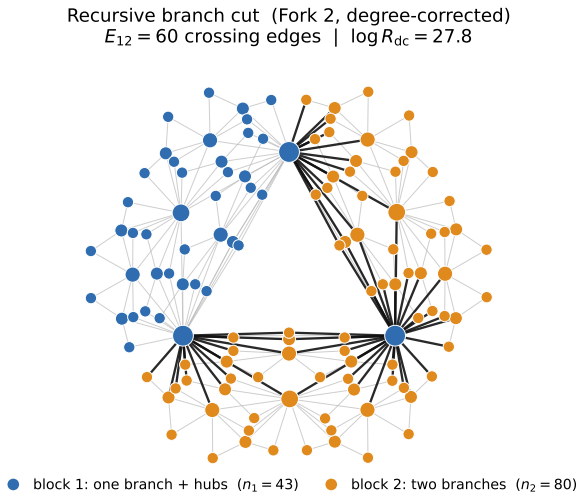

FIG. 2. The recursive branch partition of the pseudofractal web (generation $T = 4$, $n = 123$; node area $\propto$ degree). One descent branch together with the three seed hubs forms block 1; the other two branches form block 2. The two blocks are joined by only $E_{12} = 60$ crossing edges (highlighted), a fraction $\sim (2/3)^T$ of all edges. This is the closed-form partition evaluated by both detection rules, and the one the degree-corrected model favours.

unpartitioned network. Under a Bernoulli SBM with within/within/between connection probabilities $\theta_1, \theta_2, \theta_{12}$ and a Bernoulli(π) label prior, the marginal likelihood (evidence) of a partition into blocks of sizes (n_1, n_2) is

$$P = \frac{1}{Z} \int \Big[\prod_{i\in\{1,2\}} \theta_i^{E_i} (1-\theta_i)^{\binom{n_i}{2}-E_i} \Big] \times \theta_{12}^{E_{12}} (1-\theta_{12})^{n_1 n_2 - E_{12}} \times \pi^{n_1} (1-\pi)^{n_2} \, P(\theta)\, P(\pi)\, d\theta\, d\pi, \tag{5}$$

with symmetric priors $P(\theta_j) = \mathrm{Beta}(\theta_j; \alpha, \alpha)$ ($j \in \{1, 2, 12\}$) and $P(\pi) = \mathrm{Beta}(\pi; \alpha, \alpha)$. Each one-dimensional integral collapses to a Beta function through $\int_0^1 x^{E+\alpha-1}(1-x)^{M-E+\alpha-1} dx = \mathrm{B}(E+\alpha,\, M-E+\alpha)$. A block that carries no data integrates to a bare prior normaliser $\mathrm{B}(\alpha, \alpha)$ that does not cancel, a Bayesian Occam penalty for unused parameters. The unpartitioned network is the special case $n_1 = 0$: its two empty blocks each leave one $\mathrm{B}(\alpha, \alpha)$, its single occupied block sees all E_T edges among $\binom{n_T}{2}$ pairs, and its label integral is $\mathrm{B}(\alpha, n_T+\alpha)$.

III. PLAIN EVIDENCE RATIO AND ITS TRANSITION

The global constant Z and the common prefactor cancel in the ratio $R \equiv P_{\mathrm{split}}/P_{\mathrm{null}}$ of the branch partition to the unpartitioned network,

$$R = \frac{\mathrm{B}\big(E_1+\alpha,\, \binom{n_1}{2}-E_1+\alpha\big)\, \mathrm{B}\big(E_2+\alpha,\, \binom{n_2}{2}-E_2+\alpha\big)\, \mathrm{B}\big(E_{12}+\alpha,\, n_1 n_2 - E_{12}+\alpha\big)\, \mathrm{B}\big(n_1+\alpha,\, n_2+\alpha\big)}{\mathrm{B}(\alpha,\alpha)^2\, \mathrm{B}\big(E_T+\alpha,\, \binom{n_T}{2}-E_T+\alpha\big)\, \mathrm{B}\big(\alpha,\, n_T+\alpha\big)}, \tag{6}$$

with the counts (2)–(4) inserted. The posterior weight of the partition, restricted to these two hypotheses, is

$$P(\mathrm{split}) = \frac{R}{1+R} = \frac{1}{1+e^{-\log R}}. \tag{7}$$

The branch cut is *unbalanced* ($n_1 \neq n_2$), so the label factor $\mathrm{B}(n_1+\alpha, n_2+\alpha)$ carries a genuine cost; it is precisely this term that supplies a transition. Evaluated in logarithmic form with 80 digit arithmetic, $\log R$ is negative and decreasing for small generations and then turns sharply positive, Table I: the web is *not* partitioned while small, and flips to preferring the partition at generation $T = 6$, $n = 1095$.

The asymptotics are obtained from the counts (2)–(4) with Stirling's approximation. Writing $x \equiv 3^T$, every block is sparse ($E \sim x$ edges among $\sim x^2$ pairs), the seam $E_{12} \sim 2^T$ and the null label are subleading, and the $x \log x$ and x terms cancel between split and null, leaving

$$\log R \;\sim\; \Big(\ln 3 - \tfrac{2}{3}\ln 2\Big)\, n \;\approx\; 0.6365\, n, \tag{8}$$

T	n	$\log R$ (α=0.5)	$\log R$ (α=1)	$\log R$ (α=2)	P(split)
3	42	−19.01	−17.33	−16.01	≈ 0
4	123	−37.54	−36.29	−35.90	≈ 0
5	366	−52.72	−52.08	−52.91	≈ 0
6	1095	15.20	15.17	12.98	≈ 1
7	3282	506.25	505.52	501.90	≈ 1
8	9843	2667.04	2665.58	2660.51	≈ 1

TABLE I. Plain Bernoulli SBM: log-evidence ratio (6) for the recursive branch partition of the pseudofractal web. The sign changes between $T = 5$ and $T = 6$; P(split) is for $\alpha = 1$.

linear in n and independent of the prior α. The same scaling was obtained for the partition of the ring [**?**]. A numerical fit of the discrete slope $\Delta \log R/\Delta n$ converges to 0.636514, matching Eq. (8) to six digits, with the residual falling as $(2/3)^T$.

IV. DEGREE-CORRECTED EVIDENCE AND THE CONFIGURATION-MODEL NULL

Because the web is scale free, the plain transition of Sec. III is partly an artefact: a block model gains evidence simply by giving the hub-rich and hub-poor regions different densities. To isolate genuine community structure I replace the null by the configuration model, which fixes the degree sequence, and the detection rule by a degree-corrected SBM.

A. Construction

Model the (sparse) graph as a Poisson multigraph with degree-corrected rates $\langle A_{ij}\rangle = \theta_i\theta_j\,\omega_{g_ig_j}$ and $\theta_i = k_i/\sqrt{2m}$, where $2m = \sum_i k_i = 2E_T$ and g_i is the block of i [7]. With this choice $\omega_{rs} = 1$ reproduces the configuration model exactly, with expected block-pair edge counts

$$\Omega_{rr} = \frac{\kappa_r^2}{4m}, \qquad \Omega_{rs} = \frac{\kappa_r\kappa_s}{2m}\ (r \neq s), \qquad \sum_{r\le s}\Omega_{rs} = m, \tag{9}$$

where $\kappa_r = \sum_{i\in r} k_i$ is the degree sum of block r. The Poisson likelihood of the graph factorises as

$$P(A \mid \omega, g) = C(A, \boldsymbol{k}) \prod_{r\le s} \omega_{rs}^{e_{rs}}\, e^{-\omega_{rs}\Omega_{rs}}, \tag{10}$$

where e_{rs} is the observed number of edges between blocks r, s, and the prefactor $C(A, \boldsymbol{k})$ depends only on the (fixed) degrees and adjacency, not on the partition g; it therefore *cancels* in any evidence ratio between partitions. Placing a symmetric conjugate prior $\omega_{rs} \sim \mathrm{Gamma}(\alpha, \alpha)$ (mean 1, centred on the configuration model) and integrating,

$$\begin{aligned} Z(e, \Omega) &= \frac{\alpha^\alpha}{\Gamma(\alpha)} \int_0^\infty \omega^{e+\alpha-1} e^{-(\Omega+\alpha)\omega}\, d\omega \\ &= \frac{\alpha^\alpha}{\Gamma(\alpha)}\, \frac{\Gamma(e+\alpha)}{(\Omega+\alpha)^{e+\alpha}}. \end{aligned} \tag{11}$$

The degree-corrected evidence ratio of the two-block partition to the one-block configuration null is

$$R_{\mathrm{dc}} = \frac{Z(E_1, \Omega_{11})\, Z(E_2, \Omega_{22})\, Z(E_{12}, \Omega_{12})}{Z(E_T, \Omega_{\mathrm{tot}})}, \tag{12}$$

$$\Omega_{\mathrm{tot}} = E_T, \tag{13}$$

carrying an Occam factor $[\alpha^\alpha/\Gamma(\alpha)]^2$ for its two extra affinities, exactly parallel to the $\mathrm{B}(\alpha, \alpha)^2$ penalty of the plain model. For large arguments Eq. (12) reduces, term by term, to the degree-corrected assortativity signal

$$\log R_{\mathrm{dc}} \sim \sum_{r\le s} e_{rs} \ln \frac{e_{rs}}{\Omega_{rs}}, \tag{14}$$

the Kullback–Leibler surprise of the observed block-pair counts relative to configuration-model expectations—the Bayesian, degree-corrected counterpart of modularity [9].

T	n	$\log R_{\mathrm{dc}}$ (α=0.5)	$\log R_{\mathrm{dc}}$ (α=1)	$\log R_{\mathrm{dc}}$ (α=2)
2	15	−2.52	−1.77	−1.16
3	42	0.02	0.77	1.37
4	123	27.20	27.84	28.22
5	366	159.45	159.92	159.95
6	1095	682.44	682.70	682.32
7	3282	2554.18	2554.23	2553.42

TABLE II. Degree-corrected (configuration-null) log-evidence ratio (12) for the same branch partition. The sign changes between $T = 2$ and $T = 3$: the partition is preferred at far smaller sizes than under the plain model, Table I.

B. The partition survives degree correction

For the branch cut the block degree sums are again closed form,

$$\kappa_1 = 2\cdot 3^T + 2^{T+2}, \qquad \kappa_2 = 4\cdot 3^T - 2^{T+2}, \tag{15}$$

with $\kappa_1 + \kappa_2 = 2E_T$. Inserting (15) into Eq. (9) gives, in the large-T limit, within-block densities that *exceed* the configuration-model expectation while the crossing density vanishes:

$$\frac{E_1}{\Omega_{11}} \to 3, \qquad \frac{E_2}{\Omega_{22}} \to \tfrac{3}{2}, \qquad \frac{E_{12}}{\Omega_{12}} \sim \left(\tfrac{2}{3}\right)^T \to 0. \tag{16}$$

Both branches hold more internal edges than their degrees alone would place, and are joined by exponentially fewer crossing edges than expected: this is assortative community structure that is not reducible to the degree sequence. Consequently $\log R_{\mathrm{dc}} > 0$ already from $T = 3$ ($n = 42$), Table II, and grows as

$$\log R_{\mathrm{dc}} \sim \left(2\ln 3 - \tfrac{4}{3}\ln 2\right) n \approx 1.2730\, n, \tag{17}$$

again prior independent, and *exactly twice* the plain slope (8), since $2\ln 3 - \frac{4}{3}\ln 2 = 2(\ln 3 - \frac{2}{3}\ln 2)$. The leading term follows from Eq. (14) and the ratios (16): $E_1 \ln 3 + E_2 \ln\frac{3}{2} \to 3^T(3\ln 3 - 2\ln 2) = (2\ln 3 - \frac{4}{3}\ln 2)\, n$. The numerical slope converges to 1.273028, confirming Eq. (17).

C. Degree correction reverses the preferred cut

The two rules disagree about *which* partition is best. Alongside the recursive branch cut, consider the hub–leaf (core–periphery) cut that a plain block model most prefers, obtained by thresholding on degree (Fig. 3). Table III contrasts the two cuts under the two nulls.

The plain model rates the hub–leaf cut far above the branch cut; degree correction inverts this, rating the recursive branch cut well above the hub–leaf one. In other words, the large plain evidence of the hub–leaf cut is mostly the degree gradient, which the configuration null absorbs, whereas the branch cut encodes structure that

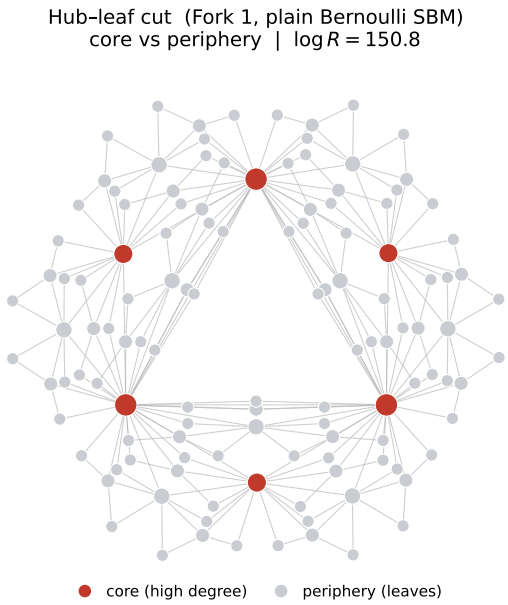

FIG. 3. The hub–leaf (core–periphery) partition of the same web ($T = 4$), separating the highest-degree vertices (core) from the low-degree leaves (periphery). The plain Bernoulli SBM rates this cut far above the recursive one of Fig. 2, but it merely tracks the degree gradient: the configuration-model null absorbs it, and the degree-corrected model demotes it (Table III).

T	n	cut	plain $\log R$	$\log R_{\rm dc}$
5	366	branch	−52.08	159.92
5	366	hub–leaf	508.74	19.32
7	3282	branch	505.52	2554.23
7	3282	hub–leaf	4844.88	209.25

TABLE III. Which cut wins depends on the null ($\alpha = 1$). The plain SBM prefers the hub–leaf cut (a degree artefact); the degree-corrected model prefers the recursive branch cut (genuine community structure).

persists once degrees are fixed. (A local search over all bipartitions under $R_{\rm dc}$ finds an optimum that overlaps the branch cut by $\approx 65\%$; the closed-form branch cut is thus a tractable witness of the degree-corrected structure, not necessarily its global optimum—mirroring the role of the balanced arc for the ring.)

V. HIERARCHICAL COMMUNITY STRUCTURE

The branch cut of Sec. IV lumps two of the three descent branches into one block, so the thin seam between them is misread as within-block structure. Separating all three branches should therefore fit better. Placing each branch with one seed hub in its own block gives a symmetric three-community partition whose counts are again

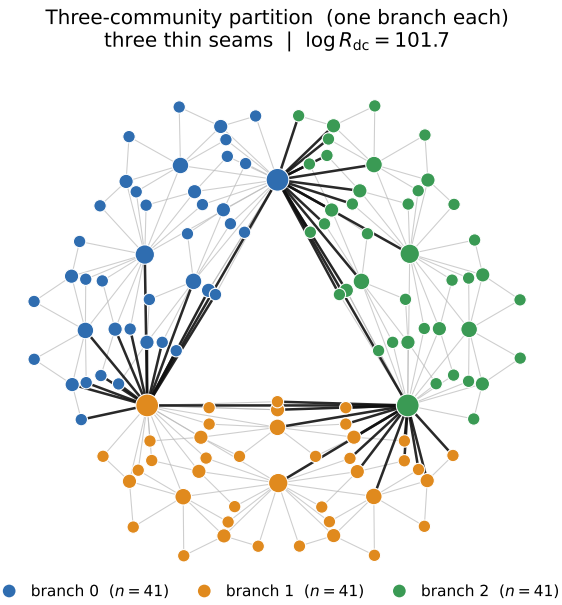

FIG. 4. The symmetric three-community partition of the web ($T = 4$), one descent branch per block. All three within-block densities exceed the configuration-model expectation and all three seams (highlighted) vanish as $(2/3)^T$, so the degree-corrected evidence, Eq. (20), beats the two-block cut of Fig. 2.

closed form (verified through $T = 8$):

$$n_r = \tfrac{1}{2}(3^T{+}1), \quad \kappa_r = 2{\cdot}3^T, \quad e_{rr} = 3^T{-}2^T, \quad e_{rs} = 2^T, \tag{18}$$

for each block r and each pair $r \neq s$. With $\Omega_{rr} = 3^T/3$ and $\Omega_{rs} = 2{\cdot}3^T/3$ from Eq. (9), every block is assortative and every seam vanishing,

$$\frac{e_{rr}}{\Omega_{rr}} = 3\big(1 - (2/3)^T\big) \to 3, \qquad \frac{e_{rs}}{\Omega_{rs}} = \tfrac{3}{2}(2/3)^T \to 0, \tag{19}$$

a genuine three-community structure. Its degree-corrected evidence exceeds the two-block value at every size (Fig. 4) and grows as

$$\log R_{\rm dc}^{(3)} \sim 2\ln 3\, n \approx 2.197\, n, \tag{20}$$

larger than the two-block slope (17) by exactly $\frac{4}{3}\ln 2\, n$—the evidence the branch cut forfeited by merging two communities.

The construction is self-similar, so the refinement does not stop at three. Each branch is itself a pseudofractal web grown from an edge, and removing the generation-d edge set partitions the graph into 3^{d+1} statistically identical sub-webs. The degree-corrected evidence keeps rising as the resolution is refined (Table IV); at fixed level d it grows as

$$\log R_{\rm dc}^{(d)} \sim 2(d{+}1)\ln 3\, n, \tag{21}$$

so each additional level adds a further $\simeq 2\ln 3\, n$. Deeper partitions win until the blocks shrink to where the seams

T	n	K=3	K=9	K=27	K=81	$K_{\rm opt}$
4	123	102	164	88	−37	9
5	366	436	826	796	360	9
6	1095	1601	3190	3970	2939	27
7	3282	5458	11013	15334	15042	27
8	9843	17823	36005	52453	60558	81
9	29526	56658	114244	169391	211877	81

TABLE IV. Degree-corrected log-evidence $\log R_{\rm dc}$ ($\alpha = 1$) for the hierarchical partition into $K = 3^{d+1}$ communities (K=3: the symmetric partition of Eq. (18); deeper levels: each generation-d sub-web, with shared vertices attached to the majority neighbouring block). The maximum in each row (column $K_{\rm opt}$) shifts to finer resolution as the web grows, tracking $K_{\rm opt} = 3^{\lfloor T/2 \rfloor} \sim \sqrt{n}$, Eq. (22).

and the Occam penalty are no longer negligible. Over the accessible range $T \leq 9$ the optimum sits at

$$K_{\rm opt} = 3^{\lfloor T/2 \rfloor} \sim \sqrt{n}, \tag{22}$$

so the web is best described by of order $\sqrt{n}$ communities of $\sim \sqrt{n}$ nodes each—the geometric midpoint of its own hierarchy. The single branch cut of Sec. IV is merely the top of this tree; the full description is the nested, hierarchical block structure characteristic of self-similar graphs [10].

VI. THE RAMSEY COMMUNITY NUMBER

Following [5], the Ramsey community number is the minimum size at which the connectivity is preferentially described by communities to a prescribed certainty q,

$$r_\kappa(q) = \min\{\, n_T : \; P(\text{split}) \geq q \,\}, \tag{23}$$

with P(split) from the relevant evidence ratio. Because the web exists only at the discrete sizes n_T of Eq. (1), r_κ takes one of those values. The results are collected in Table V. Under the plain model the transition is so abrupt—$\log R$ leaps from -52 at $T = 5$ to $+15$ at $T = 6$ to $+506$ at $T = 7$—that $r_\kappa = 1095$ for every certainty level up to $q \simeq 1 - 10^{-5}$ and every prior α, a step sharper than the ring's gradual crossover. Under degree correction the web breaks at $r_\kappa = 42$ ($q = \frac{1}{2}$) or 123 ($q \geq 0.9$), again essentially α independent.

$$r_\kappa^{\rm plain} = 1095, \qquad r_\kappa^{\rm dc} = 42\text{–}123. \tag{24}$$

The two numbers make the interpretation explicit. The plain $r_\kappa = 1095$ conflates two effects and switches on only when the accumulated degree-plus-community signal wins; the degree-corrected $r_\kappa = 42$ measures the community signal alone, which is present almost from the start. That the degree-corrected threshold is the *smaller* of the two shows the recursion imprints community structure early and the degree gradient, if anything, delays the plain detector rather than driving it.

model	α	$r_\kappa^{0.5}$	$r_\kappa^{0.9}$	$r_\kappa^{0.99}$	$r_\kappa^{0.999}$
plain	0.5	1095	1095	1095	1095
plain	1.0	1095	1095	1095	1095
plain	2.0	1095	1095	1095	1095
dc	0.5	42	123	123	123
dc	1.0	42	123	123	123
dc	2.0	42	123	123	123

TABLE V. Ramsey community number r_κ^q (smallest generation size n_T with $P(\text{split}) \geq q$) for the pseudofractal web, under the plain Bernoulli SBM and the degree-corrected (dc) configuration-null model. Degree correction lowers r_κ by more than an order of magnitude and is nearly prior independent.

VII. CONCLUSIONS

I have computed the Ramsey community number of the deterministic pseudofractal scale-free web analytically, under two detection rules whose Bayesian evidences are both closed form: a plain Bernoulli SBM and a degree-corrected Poisson SBM tested against the configuration-model null. The natural recursive bipartition of the web has block sizes and edge counts known exactly at every generation, Eqs. (2)–(4) and (15), which makes both evidence ratios explicit.

The results are fivefold. (1) Under the plain SBM the web is unpartitioned while small and acquires a finite Ramsey community number $r_\kappa = 1095$, above which $\log R \sim (\ln 3 - \frac{2}{3}\ln 2)n$, prior independent. (2) Under degree correction the same partition survives—its within-block densities exceed, and its crossing density falls exponentially below, configuration-model expectations, Eq. (16)—and breaks at the much smaller $r_\kappa = 42$, with $\log R_{\rm dc} \sim (2\ln 3 - \frac{4}{3}\ln 2)n$, exactly twice the plain slope. (3) Degree correction reverses the ordering of the candidate cuts, demoting the hub–leaf split favoured by the plain model and elevating the recursive one, so the effect is genuine community structure, not degree heterogeneity. (4) Both transitions are prior independent in their asymptotic slope, and the discrete generation sizes make r_κ a sharp step. (5) Because the web is self-similar, the degree-corrected evidence keeps rising as the partition is refined—each hierarchical level adding $\simeq 2\ln 3\, n$, Eq. (21)—so the optimal description is not two or three communities but a nested hierarchy of $K_{\rm opt} = 3^{\lfloor T/2 \rfloor} \sim \sqrt{n}$ blocks.

Together with the companion ring calculation [**?**], these results extend the exactly solvable theory of community emergence from a homogeneous graph to a strongly heterogeneous one, and supply the degree-correction control that a scale-free setting demands. They confirm that a purely local recursive rule can build hierarchical communities over and above the degree sequence it also produces. Natural extensions include a proof that $K_{\rm opt} \sim \sqrt{n}$ asymptotically together with the exact evidence at the optimal resolution, other deterministic fractals such as the (u, v)-flowers, and the scaling of

r_κ with the branching number—each within reach of the present closed-form approach.

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